DevOps Course Study Guide

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Section 1: DevOps Foundations

- DevOps Lifecycle & Culture
- Agile, Scrum & SDLC
- Delivery Metrics & SRE Basics

Section 2: Linux & Networking

- Linux Administration
- Shell Scripting
- DNS, HTTP, SSH & Troubleshooting

Section 3: Git & Collaboration

- Git Fundamentals
- Branching & Pull Requests
- GitHub Workflow & Releases

Section 4: Docker & Containers

- Images & Containers
- Dockerfile & Registries
- Docker Compose & Networking

Section 5: CI/CD Automation

- Pipeline Design
- GitHub Actions & Jenkins
- Testing, Artifacts & Deployment

Section 6: Cloud & Infrastructure

- AWS Core Services
- Terraform Infrastructure as Code
- Ansible Configuration Management

Section 7: Kubernetes

- Pods, Deployments & Services
- ConfigMaps, Secrets & Storage
- Helm, Scaling & Operations

Section 8: Observability & DevSecOps

- Prometheus & Grafana
- Logging, Alerting & Incident Response
- Security Scanning & Final Project